AS Chemistry homework · Paper 2
σ/π bonds, hybridisation and bond energy

CIE 9701 AS Chemistry · Paper 2 of 2 · 20 marks · about 40 minutes

Topic: 3.4 LO2 σ and π bonds and hybridisation (sp, sp^2, sp^3); 3.4 LO3 bond energy and comparing reactivity of covalent molecules. Answer in English. Section A is Paper 1 style (1 mark each). Section B is written; use CIE language (orbital overlap, σ/π, hybridisation, bond energy).

Section	Content	Marks
A	Multiple choice	8
B1	2024 MJ Paper 23 Question 1(a)(ii)	4
B2	2024 MJ Paper 21 Question 5(b)(i)	1
B3	2024 MJ Paper 21 Question 5(b)(ii)	2
B4	2022 ON Paper 22 Question 1(c)(iii)	2
B5	2021 MJ Paper 21 Question 2(b)	1
B6	2025 MJ Paper 24 Question 3(d)(ii)	2
	Total	**20**

Section A — Multiple-choice questions (8 marks)

A1. [1]

The bonding between two atoms of nitrogen in an N_2 molecule involves the hybridisation of atomic orbitals to form *sp* orbitals. Which row is correct?

	formation of the σ bond between the nitrogen atoms in N_2	type of orbital which contains the lone pair of electrons on each nitrogen atom in N_2
A	an *sp* orbital from one atom overlaps with an *sp* orbital of the other atom	*p*
B	an *sp* orbital from one atom overlaps with an *sp* orbital of the other atom	*sp*
C	an *s* orbital from one atom overlaps with a *p* orbital of the other atom	*p*
D	an *s* orbital from one atom overlaps with a *p* orbital of the other atom	*sp*

2025 ON Paper 12 Question 5

A2. [1]

Propene, hydrogen cyanide and carbon dioxide each contain π bonds.

Which molecules contain two π bonds?

A carbon dioxide and hydrogen cyanide
B carbon dioxide and propene
C hydrogen cyanide and propene
D hydrogen cyanide only

2025 FM Paper 12 Question 16

A3. [1]

How many σ bonds are present in one $H{-}C{\equiv}C{-}C(CH_3){=}CH(CH_3)$ molecule?

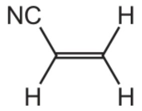
A 5 **B** 11 **C** 13 **D** 16

2022 ON Paper 11 Question 5

A4. [1]

26 The diagram shows the structure of X.

X



Which row is correct?

	number of σ bonds in X	type of hybridisation of the carbon atoms in X
A	4	sp and sp^2
B	4	sp^2 and sp^3
C	6	sp and sp^2
D	6	sp^2 and sp^3

2024 MJ Paper 11 Question 26

A5. [1]

What is the total number of sp^3 hybridised atomic orbitals used in the bonding of but-2-ene?

A 2 **B** 4 **C** 6 **D** 8

2025 FM Paper 12 Question 39

A6. [1]

NH_3 and HCN react together to form NH_4CN, an ionic compound.

Which row states the number of coordinate bonds and the number of π bonds in one formula unit of NH_4CN?

	number of coordinate bonds	number of π bonds
A	0	2
B	1	2
C	0	3
D	1	3

2025 ON Paper 11 Question 6

A7. [1]

Ammonium ions, $NH_4{}^+$, are formed when ammonia gas reacts with hydrogen chloride gas.

Which statement about the changes that occur in this reaction is correct?

A The dipole moment of an ammonium ion is greater than the dipole moment of an ammonia molecule.
B The H–N–H bond angle decreases when an ammonium ion is formed.
C The hybridisation of nitrogen does **not** change.
D There is electron transfer from nitrogen to chlorine.

2023 ON Paper 12 Question 5

A8. [1]

Which equation represents an enthalpy change that is the average bond energy of the C–H bond in methane?

A $\frac{1}{4}\mathrm{C}(g) + \mathrm{H}(g) \rightarrow \frac{1}{4}\mathrm{CH_4}(g)$
B $\frac{1}{4}\mathrm{CH_4}(g) \rightarrow \frac{1}{4}\mathrm{C}(g) + \mathrm{H}(g)$
C $\mathrm{CH_4}(g) \rightarrow \mathrm{C}(g) + 4\mathrm{H}(g)$
D $\mathrm{CH_4}(g) \rightarrow \mathrm{CH_3}(g) + \mathrm{H}(g)$

2022 MJ Paper 13 Question 9

Section B — Structured questions (12 marks)

B1 · 2024 May/June Paper 23 Question 1(a)(ii) (4 marks)

Covalent bonds can be σ bonds or π bonds.

B1(a)(ii) [4]

Complete Table 1.1 to show the number of σ and π bonds in a molecule of N_2 and to describe how the orbitals overlap to form σ and π bonds.

	σ bond	π bond
number of bonds in N_2		
how the orbitals overlap		

Table 1.1

B2 and B3 · 2024 May/June Paper 21 Question 5(b) (3 marks)

Hydrocarbon molecules contain covalent bonds.

A C=C bond in an alkene is made from a σ bond and a π bond.

B2(b)(i) [1]

Identify the hybridisation of the carbon atoms in a C=C bond in an alkene.

B3(b)(ii) [2]

Draw labelled diagrams to show, in terms of orbital overlap, how the σ and π bonds are made in a C=C bond.

σ bond

π bond

B4 · 2022 October/November Paper 22 Question 1(c)(iii) (2 marks)

B4(c)(iii) [2]

1 Species such as NH_4^+, CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

(c) NH_4^+ is a Brønsted–Lowry acid.

(iii) The nitrogen atom in NH_4^+ is sp^3 hybridised. sp^3 orbitals form from the mixing of one 2s and three 2p orbitals.

Sketch the shapes of a 2s and a $2p_x$ orbital on the axes in Fig. 1.1.

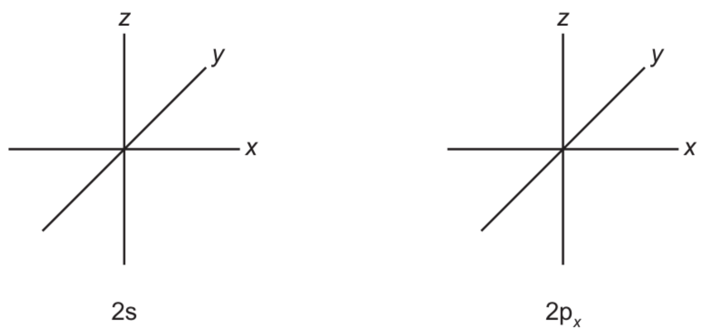


Fig. 1.1

[2]

2022 ON Paper 22 Question 1(c)(iii)

B5 · 2021 May/June Paper 21 Question 2(b) (1 mark)

Carbon monoxide gas, CO(g), and nitrogen gas, N_2(g), are both diatomic molecules.

B5(b) [1]

N_2(g) is less reactive than CO(g) even though N_2(g) has a lower bond energy than CO(g). Suggest why CO(g) is more reactive than N_2(g).

B6 · 2025 May/June Paper 24 Question 3(d)(ii) (2 marks)

B6(d)(ii) [2]

Describe and explain the relative thermal stabilities of the hydrogen halides HCl and HI.